

Operations Research

How to Write the Dual of Any Linear Program

A methodology guide for Lecture 4, with the reason behind every sign rule

0. The only idea you need

Every sign rule in duality comes from one trick, and if you remember the trick you can rebuild the rules on the spot.

The trick

Take a *maximisation*. We want to prove an upper bound: “no feasible x earns more than U ”. The only tool available is: **multiply each constraint by a weight, add the weighted constraints, and compare the result with the objective.**

For this to produce a valid bound, two things must hold:

- (I) every weighted row must still point the right way ($\leq$, since we want an *upper* bound);
- (II) the weighted sum must be *at least* the objective, for every allowed x .

Requirement (I) decides the **sign of each dual variable**. Requirement (II) decides the **direction of each dual constraint**. Nothing else is involved.

Let us see the trick once in full on the lecture’s example, then extract the rules.

Lecture 4 example, done slowly

$$\max 2x_1 + 3x_2 \quad \text{s.t.} \quad 4x_1 + 8x_2 \leq 12, \quad 2x_1 + x_2 \leq 3, \quad 3x_1 + 2x_2 \leq 4, \quad x \geq 0.$$

Pick weights y_1, y_2, y_3 and multiply:

$$y_1(4x_1 + 8x_2) \leq 12y_1, \quad y_2(2x_1 + x_2) \leq 3y_2, \quad y_3(3x_1 + 2x_2) \leq 4y_3.$$

(I) These three lines are only correct if $y_1, y_2, y_3 \geq 0$: multiplying $\leq$ by a negative number would flip it. Add them and regroup by variable:

$$\underbrace{(4y_1 + 2y_2 + 3y_3)}_{d_1} x_1 + \underbrace{(8y_1 + y_2 + 2y_3)}_{d_2} x_2 \leq 12y_1 + 3y_2 + 4y_3.$$

(II) We want the left side to be at least $2x_1 + 3x_2$. Since $x_1, x_2 \geq 0$, it is enough that each coefficient dominates: $d_1 \geq 2$ and $d_2 \geq 3$. Then

$$2x_1 + 3x_2 \leq d_1x_1 + d_2x_2 \leq 12y_1 + 3y_2 + 4y_3.$$

The best bound is the smallest right-hand side. So: $\min 12y_1 + 3y_2 + 4y_3$ s.t. $4y_1 + 2y_2 + 3y_3 \geq 2, 8y_1 + y_2 + 2y_3 \geq 3, y \geq 0$. That is the dual.

Notice what d_j is: the coefficients of x_j read *down* its column, dotted with y . In matrix form $d = A^\top y$. That is the whole reason the dual has $A^\top$.

1. Rule 1: the row direction fixes the sign of the dual variable

Why

Requirement (I): after multiplying by y_i , row i must read “ $\dots \leq \dots$ ”. Ask: *what sign of y_i makes this row point $\leq$?*

Primal row i (max problem)	Dual variable	Reason
$a_i^\top x \leq b_i$	$y_i \geq 0$	a nonnegative multiplier keeps $\leq$
$a_i^\top x \geq b_i$	$y_i \leq 0$	a nonpositive multiplier turns $\geq$ into $\leq$
$a_i^\top x = b_i$	$y_i \in \mathbb{R}$ (free)	an equality survives any multiplier

Check it on an example. Row: $x_1 - x_3 \geq -2$. Multiply by $y = -3$: $-3x_1 + 3x_3 \leq 6$. It now points $\leq$, usable. Multiply by $y = +3$ instead: $3x_1 - 3x_3 \geq -6$, which points the wrong way and is useless for an upper bound. So on a $\geq$ row the weight must be ≤ 0 .

2. Rule 2: the variable sign fixes the direction of the dual constraint

Why

Requirement (II): we need $c_j x_j \leq d_j x_j$ for every value x_j is allowed to take, where $d_j = (A^\top y)_j$ is the coefficient of x_j in the weighted sum. Compare the two *numbers* c_j and d_j , and ask what inequality between them survives multiplication by x_j .

Primal variable x_j	Dual constraint j	Reason
$x_j \geq 0$	$d_j \geq c_j$	multiplying $c_j \leq d_j$ by $x_j \geq 0$ keeps it
$x_j \leq 0$	$d_j \leq c_j$	multiplying by $x_j \leq 0$ flips it, so start from $c_j \geq d_j$
$x_j \in \mathbb{R}$ (free)	$d_j = c_j$	must work for both signs of x_j ; only equality does

Check the middle row. Suppose $x_2 \leq 0$ and we had $d_2 \geq c_2$, say $d_2 = 5$, $c_2 = 1$, $x_2 = -1$. Then $d_2 x_2 = -5$ and $c_2 x_2 = -1$, so $d_2 x_2 < c_2 x_2$: the weighted sum is *smaller* than the objective, and the bound fails. With $d_2 \leq c_2$ instead, say $d_2 = 0$, $c_2 = 1$: $d_2 x_2 = 0 \geq -1 = c_2 x_2$. Correct.

3. Rule 3: the dual objective

The right-hand side of the weighted sum is $b_1 y_1 + \dots + b_m y_m = b^\top y$. It is an upper bound on the profit for every admissible y . We want the *tightest* bound, so the dual **minimises** $b^\top y$. The b_i go in with their own signs; nothing is flipped by hand (any flipping is already encoded in the sign of y_i).

4. The recipe

Step 1 – Make it a maximisation

If the primal is a minimisation, you have two options. Either convert: $\min c^\top x$ is $-\max(-c)^\top x$, apply the rules below, and remember the sign at the end. Or use the mirrored table of Section 6 directly. Both give the same dual.

Step 2 – Rows become variables (Rule 1)

Go down the constraints. For each row i create y_i and write its sign next to it: $\leq \rightarrow y_i \geq 0$; $\geq \rightarrow y_i \leq 0$; $= \rightarrow y_i$ free. Do not include the sign constraints $x_j \geq 0$ here; they are not rows, they are handled in Step 3.

Step 3 – Columns become constraints (Rule 2)

Go across the variables. For each x_j :

- read its coefficients *down* the rows: $(a_{1j}, a_{2j}, \dots, a_{mj})$, keeping every sign, writing 0 where x_j is absent;
- write the left side $a_{1j}y_1 + a_{2j}y_2 + \dots + a_{mj}y_m$;
- write the right side c_j (the objective coefficient of x_j);
- choose the direction from the sign of x_j : $x_j \geq 0 \rightarrow \geq$; $x_j \leq 0 \rightarrow \leq$; x_j free $\rightarrow =$.

Step 4 – Objective (Rule 3)

Minimise $b_1y_1 + \dots + b_my_m$, using the right-hand sides exactly as they appear in the rows.

Step 5 – Check dimensions

m primal rows $\Rightarrow m$ dual variables. n primal variables $\Rightarrow n$ dual constraints. If the counts do not match, something was skipped.

5. Worked example with every kind of sign

$$\begin{array}{rcll}
 \max & 2x_1 & - & x_2 & + & 3x_3 & & \\
 \text{s.t.} & x_1 & + & x_2 & + & x_3 & \leq & 6 \quad \text{row 1} \\
 & x_1 & & & - & x_3 & \geq & -2 \quad \text{row 2} \\
 & 2x_1 & + & x_2 & & & = & 4 \quad \text{row 3} \\
 & x_1 \geq 0, & & x_2 \leq 0, & & x_3 \in \mathbb{R}. & &
 \end{array}$$

Step 2, rows. Row 1 is $\leq$: $y_1 \geq 0$. Row 2 is $\geq$: $y_2 \leq 0$. Row 3 is $=$: y_3 free.

Step 3, columns. Lay the coefficients out as a table so nothing is missed:

	x_1	x_2	x_3	rhs
row 1 ($y_1 \geq 0$)	1	1	1	6
row 2 ($y_2 \leq 0$)	1	0	-1	-2
row 3 (y_3 free)	2	1	0	4
objective c_j	2	-1	3	
sign of x_j	≥ 0	≤ 0	free	

Now read each column downward and attach the direction from the bottom row:

$$\begin{aligned}
 x_1 : \quad (1, 1, 2) \cdot y = y_1 + y_2 + 2y_3 &\geq 2 & (x_1 \geq 0 \Rightarrow \geq), \\
 x_2 : \quad (1, 0, 1) \cdot y = y_1 + y_3 &\leq -1 & (x_2 \leq 0 \Rightarrow \leq), \\
 x_3 : \quad (1, -1, 0) \cdot y = y_1 - y_2 &= 3 & (x_3 \text{ free} \Rightarrow =).
 \end{aligned}$$

Step 4, objective. $\min 6y_1 - 2y_2 + 4y_3$. The -2 is copied as is.

Step 5, check. 3 rows $\rightarrow$ 3 variables y_1, y_2, y_3 . 3 variables $\rightarrow$ 3 constraints. Good.

The dual.

$$\begin{aligned}
 \min \quad & 6y_1 - 2y_2 + 4y_3 \\
 \text{s.t.} \quad & y_1 + y_2 + 2y_3 \geq 2, \\
 & y_1 + y_3 \leq -1, \\
 & y_1 - y_2 = 3, \\
 & y_1 \geq 0, \quad y_2 \leq 0, \quad y_3 \in \mathbb{R}.
 \end{aligned}$$

Remark

Students often try to “fix” row 2 by multiplying it by -1 first, turning it into $-x_1 + x_3 \leq 2$ with a nonnegative y'_2 . That is also correct, and gives the same dual with $y'_2 = -y_2$: the constraint for x_1 becomes $y_1 - y'_2 + 2y_3 \geq 2$ and the objective $6y_1 + 2y'_2 + 4y_3$. Either route works. What you may *not* do is multiply by -1 and still write $y_2 \leq 0$, or keep the row as $\geq$ and write $y_2 \geq 0$. The sign of the variable and the direction of the row must be decided together, once.

6. If the primal is a minimisation

Everything mirrors. Now we want a *lower* bound on the cost, so (I) every weighted row must point $\geq$, and (II) the weighted sum must be *below* the cost. Re-deriving Rules 1 and 2 with those two changes gives:

Primal is a max		Primal is a min	
dual is a min of $b^\top y$		dual is a max of $b^\top y$	
row $\leq$	$y_i \geq 0$	row $\geq$	$y_i \geq 0$
row $\geq$	$y_i \leq 0$	row $\leq$	$y_i \leq 0$
row $=$	y_i free	row $=$	y_i free
$x_j \geq 0$	constraint $\geq c_j$	$x_j \geq 0$	constraint $\leq c_j$
$x_j \leq 0$	constraint $\leq c_j$	$x_j \leq 0$	constraint $\geq c_j$
x_j free	constraint $= c_j$	x_j free	constraint $= c_j$

One way to remember the whole table: **the “natural” row direction gets the nonnegative variable, and the “natural” variable sign gets the natural constraint direction.** For a

max problem the natural row is $\leq$ (resources) and the natural dual constraint is $\geq$ (the bound must dominate). For a min problem the natural row is $\geq$ (requirements) and the natural dual constraint is $\leq$ (the prices must not exceed the costs). Anything unnatural flips the corresponding sign; equality and free always pair up.

Example (Set 2, Problem 3). $\min 4x_A + 3x_B + 5x_C$ with rows $x_A + x_C \geq 2$, $x_B + x_C \geq 1$, and caps written as $-x_A \geq -2$ etc., all variables ≥ 0 . All rows are $\geq$ in a min: all $y_i \geq 0$. All variables are ≥ 0 in a min: all dual constraints are $\leq c_j$. Dual: $\max 2y_1 + y_2 - 2y_3 - 2y_4 - 2y_5$ s.t. $y_1 - y_3 \leq 4$, $y_2 - y_4 \leq 3$, $y_1 + y_2 - y_5 \leq 5$, $y \geq 0$.

7. The mistakes that cause the “sign mess”

Mistake	Fix
Treating $x_j \geq 0$ as a constraint row and giving it a dual variable.	Sign restrictions on x are <i>not</i> rows. They only decide the direction of dual constraint j . A primal with m rows has exactly m dual variables.
Reading coefficients across a row when writing a dual constraint.	Dual constraint j uses <i>column</i> j : the coefficients of x_j in every row, top to bottom.
Dropping the 0 when x_j is absent from a row.	Write the 0 explicitly in the column table; it keeps the positions aligned with $y_1, \dots, y_m$.
Flipping a $\geq$ row to $\leq$ and also writing $y_i \leq 0$.	Decide once: either keep the row and take $y_i \leq 0$, or flip the row and take $y'_i \geq 0$. Not both.
Changing the sign of b_i in the dual objective “because the row was $\geq$ ”.	Copy b_i as it stands. The row direction is fully accounted for by the sign of y_i .
Using the max table on a min problem.	Use the mirrored column of Section 6, or convert to a max first.
Forgetting that free variables give equalities.	x_j free $\Rightarrow$ dual constraint j is $=$. Equality row $\Rightarrow y_i$ free. Always paired.

8. Self-test

Write the dual of each; answers below. Do them with the column table, not from memory.

- (1) $\max x_1 + 2x_2$ s.t. $x_1 - x_2 \leq 1$, $x_1 + x_2 = 3$, $x_1 \geq 0$, x_2 free.
- (2) $\min 3x_1 + x_2 + 2x_3$ s.t. $x_1 + x_2 \geq 2$, $x_2 - x_3 \leq 1$, $x_1 + x_3 = 4$, $x_1, x_2 \geq 0$, $x_3 \leq 0$.
- (3) $\max -x_1 + 4x_2$ s.t. $2x_1 + x_2 \geq 5$, $x_1 - 3x_2 \leq -1$, $x_1 \leq 0$, $x_2 \geq 0$.

Answers.

- (1) Rows: $y_1 \geq 0$, y_2 free. Columns: $x_1 = (1, 1)$, $x_1 \geq 0$: $y_1 + y_2 \geq 1$. $x_2 = (-1, 1)$, free: $-y_1 + y_2 = 2$. Dual: $\min y_1 + 3y_2$ s.t. $y_1 + y_2 \geq 1$, $-y_1 + y_2 = 2$, $y_1 \geq 0$, y_2 free.
- (2) Min problem, use the right column of Section 6. Rows: $\geq \rightarrow y_1 \geq 0$; $\leq \rightarrow y_2 \leq 0$; $= \rightarrow y_3$ free. Columns: $x_1 = (1, 0, 1)$, $x_1 \geq 0 \rightarrow \leq$: $y_1 + y_3 \leq 3$. $x_2 = (1, 1, 0)$, $x_2 \geq 0 \rightarrow \leq$: $y_1 + y_2 \leq 1$. $x_3 = (0, -1, 1)$, $x_3 \leq 0 \rightarrow \geq$: $-y_2 + y_3 \geq 2$. Dual: $\max 2y_1 + y_2 + 4y_3$ s.t. those three, $y_1 \geq 0$, $y_2 \leq 0$, y_3 free.

- (3) Rows: $\geq \rightarrow y_1 \leq 0$; $\leq \rightarrow y_2 \geq 0$. Columns: $x_1 = (2, 1)$, $x_1 \leq 0 \rightarrow \leq$: $2y_1 + y_2 \leq -1$.
 $x_2 = (1, -3)$, $x_2 \geq 0 \rightarrow \geq$: $y_1 - 3y_2 \geq 4$. Dual: $\min 5y_1 - y_2$ s.t. $2y_1 + y_2 \leq -1$, $y_1 - 3y_2 \geq 4$,
 $y_1 \leq 0$, $y_2 \geq 0$.

Remark

If you only remember one sentence: *rows become variables and the row direction gives the variable's sign; columns become constraints and the variable's sign gives the constraint's direction; the objective is $b^\top y$ with the b_i copied as they are.* And if you forget a rule, redo the trick of Section 0 for that row or column; it takes thirty seconds and cannot be wrong.